

```
import customtkinter as ctk
from tkinter import ttk, messagebox
from database import Database
```

```
class TopicManager:
```

```
    def __init__(self, root):
```

```
        self.root = root
```

```
        self.db = Database()
```

```
        self.create_topic_ui()
```

```
    # ----- UI ----- #
```

```
    def create_topic_ui(self):
```

```
        self.clear_window()
```

```
        heading = ctk.CTkLabel(
            self.root,
            text="Topic Manager",
            font=("Arial", 28, "bold")
        )
```

```
        heading.pack(pady=20)
```

```
        # Topic Name
```

```
        self.topic_entry = ctk.CTkEntry(
            self.root,
            placeholder_text="Topic Name",
            width=250
        )
```

```
)
```

```
self.topic_entry.pack(pady=10)
```

```
# Subject
```

```
self.subject_entry = ctk.CTkEntry(  
    self.root,  
    placeholder_text="Subject",  
    width=250
```

```
)
```

```
self.subject_entry.pack(pady=10)
```

```
# Priority
```

```
self.priority_option = ctk.CTkOptionMenu(  
    self.root,  
    values=["Low", "Medium", "High"]
```

```
)
```

```
self.priority_option.pack(pady=10)
```

```
# Status
```

```
self.status_option = ctk.CTkOptionMenu(  
    self.root,  
    values=["Pending", "Completed"]
```

```
)
```

```
self.status_option.pack(pady=10)
```

```
# Add Button
```

```
add_btn = ctk.CTkButton(  
    self.root,  
    text="Add Topic",
```

```
        command=self.add_topic
    )
```

```
add_btn.pack(pady=20)
```

```
# Topics Table
```

```
self.tree = ttk.Treeview(
    self.root,
    columns=("ID", "Topic", "Subject", "Priority", "Status"),
    show="headings",
    height=10
)
```

```
self.tree.heading("ID", text="ID")
self.tree.heading("Topic", text="Topic")
self.tree.heading("Subject", text="Subject")
self.tree.heading("Priority", text="Priority")
self.tree.heading("Status", text="Status")
```

```
self.tree.pack(pady=20)
```

```
# Delete Button
```

```
delete_btn = ctk.CTkButton(
    self.root,
    text="Delete Selected Topic",
    command=self.delete_topic
)
```

```
delete_btn.pack(pady=10)
```

```
self.load_topics()
```

```
# ----- ADD TOPIC ----- #
```

```
def add_topic(self):
```

```
    topic = self.topic_entry.get()
    subject = self.subject_entry.get()
    priority = self.priority_option.get()
    status = self.status_option.get()
```

```
    if topic == "" or subject == "":
        messagebox.showerror(
            "Error",
            "All fields are required"
        )
    return
```

```
    self.db.add_topic(
        topic,
        subject,
        priority,
        status
    )
```

```
    messagebox.showinfo(
        "Success",
        "Topic Added Successfully"
    )
```

```
    self.load_topics()
```

```
# ----- LOAD TOPICS ----- #
```

```
def load_topics(self):
```

```
for row in self.tree.get_children():
    self.tree.delete(row)
```

```
topics = self.db.get_topics()
```

```
for topic in topics:
```

```
    self.tree.insert(
        "",
        "end",
        values=topic
    )
```

```
# ----- DELETE TOPIC ----- #
```

```
def delete_topic(self):
```

```
    selected = self.tree.selection()
```

```
    if not selected:
```

```
        messagebox.showerror(
            "Error",
            "Select a topic first"
        )
        return
```

```
    topic_data = self.tree.item(selected[0])
```

```
    topic_id = topic_data["values"][0]
```

```
    self.db.delete_topic(topic_id)
```

```
    messagebox.showinfo(
```

```
"Success",  
"Topic Deleted Successfully"  
)
```

```
self.load_topics()
```

```
# ----- CLEAR WINDOW ----- #
```

```
def clear_window(self):
```

```
    for widget in self.root.winfo_children():  
        widget.destroy()
```